In-class quiz-Ferroelectric ceramics

1. A Ca^{2+} cation and an O^{2-} anion are separated by a distance of 2.4 Å. Calculate the resultant dipole moment. (Charge on an electron = 1.6×10^{-19} C)

2. Consider a capacitor in a computer power supply, possessing a capacitance of 2200 μF. If a voltage of 10 V is applied to this capacitor, what will the charge on the positive plate be?

3. Which best describes the behaviour of the dielectric loss as the frequency is increased?
   a It gradually decreases.
   b It gradually increases.
   c It shows a dip around the relaxation frequency of the orientational mechanism.
   d It shows a peak around the relaxation frequency of the orientational mechanism.

4. You need to make a capacitor that will operate at low electric field strengths and store a large quantity of charge. Energy efficiency does not need to be high (i.e. loss can be tolerated). Which of the following are you most likely to place between the capacitor plates?

   a Nothing - I would use an empty capacitor.
   b A ceramic with a non-centrosymmetric structure.
   c A polymer with polar chains and an amorphous structure.
   d A polymer with polar chains and a crystalline structure.

5. What symmetry element must be absent for a material to be ferroelectric?
   a An axis of rotation.
   b A mirror plane.
   c A centre of symmetry.
   d An improper axis of rotation.

6. Which of these is not a correct definition of polarization?
   a The net dipole moment per unit volume.
   b The surface charge per unit area.
   c The movement of atoms giving rise to a dipole moment.
   d The net charge per dipole moment.

7. Why are domains in crystals found in a manner such that their polarization is coupled?

a To reduce stray field energy.
b To reduce dislocation strain energy.
c To grow the crystal in a regular manner.
d To give small grains.

8. What is needed to make ferroelectric domains useful as a binary memory store?
   a Large domains.
   b A square hysteresis loop.
   c A large coercive field.
   d Small domains.

9. If a ferroelectric, 50mm by 10 mm, has a measured surface charge of 2.5 x 10^{-4} Coulombs, and a lattice parameter of 5 x 10^{-10}, what is the dipole moment in a single cubic unit cell?
   a 6.25 x 10^{-29}
   b 6.25 x 10^{-35}
   c 1.5625 x 10^{-35}
   d 1.5625 x 10^{-29}

10. What field has to be applied to give a ferroelectric, with a square hysteresis loop, zero net polarisation?
    a The switching field.
    b The polarising field.
    c The coercive field.
    d The stray field.

11. What symmetry element must be absent for a material to be piezoelectric?
    a An axis of rotation
    b A mirror plane
    c A centre of symmetry
    d An improper axis of rotation

12. Applying a mechanical stress to a piezoelectric does not cause which of these?
    a The formation of a dipole moment
    b The movement of atoms
    c Development of polarization
    d The generation of an internal current

13. The application of an electric field to a piezoelectric does not cause which of these?
    a A change in shape of the piezoelectric

b The formation of a magnetic field
c The movement of atoms
d The formation of dipole moments to oppose the field

14. What does 'poling' do?
a Align ferroelectric dipoles to give a net polarization in a piezoelectric
b Produce a non-polarized ferroelectric
c Increase the dipole moment in each unit cell
d Produces a pole

15. Which of these will not depolarize a poled ferroelectric piezoelectric?
a Applying a very large stress to the piezoelectric
b Applying a reversed electric field to the piezoelectric
c Applying a current perpendicular to the polarization
d Applying a large amount of heat to the piezoelectric